THE MAPPING THEOREM FOR TARGETS Fib$k$

Purpose. Informal proof for §2 of `doc/pi3-s2-postnikov-handoff-7.txt` (§3c of the plan): for $P = \mathrm{Fib}\kappa$ and every Postnikov triple (A, B, k), the canonical projection $q_* : [P, \mathrm{Fib}k]_* \to [P, K(A,2)]_*$ is a bijection; and, for a class $s \in [S^2, P]_*$ constructed here, precomposition $s^* : [P, \mathrm{Fib}k]_* \to [S^2, \mathrm{Fib}k]_*$ is a bijection. Written before the formal proof; numbered statements are meant to be declarations. Foundations as in the development (univalence, images; function extensionality derived); no new axiom; the sphere axiom (`SphereAxiom`) is used in §§4–5 only.

Hypotheses. $\kappa : K(\mathbb{Z},2) \to_* K(\mathbb{Z},4)$ is any pointed map with (H4) (`KZ2Cohomology.H4`); we use the *surjectivity* of Φ_B (§3) and, through `FibreH3.FibAt[κ,h4k,B].merely-zero`, its injectivity ((H3) enters there through `H3Proof.h3`, not as a hypothesis). Instantiated with `CupSquare.kappa`, `H4Proof.h4`. *Refereed* independently (a fresh reader given the note and the library, not the conclusion; and the user): correct, with formalisation-level fixes now incorporated (e_1^{-1} is the map of the pointed equivalence `pt-inv e1`; the cancellation in Proposition 21 is on the other side, `cancel-r`; (U) for P is `merely-zero`; the connectivity of T_x inlined; the wording on lift spaces).

The idea. One lifting lemma (§2), for an arbitrary pointed source Q: if (E) every composite $F; k$ with $F : Q \to_* K(A,2)$ is merely null and (U) every $Q \to_* \Omega K(B,4)$ is merely null, then q_* is bijective on classes. It is applied twice: to $Q = P$ ((E) from (H4) and the defining null-homotopy of P; (U) is $[P, K(B,3)]_* = 0$) and to $Q = S^2$ ((E), (U): $\Omega^2 K(B,4)$ and $\Omega^3 K(B,4)$ are connected). The sphere case also produces s (no choice: s is the unique class over a chosen class $[\sigma] \in [S^2, K(\mathbb{Z},2)]_*$), and a commuting square of four maps of sets, three of them bijective, gives the fourth.

Throughout, *taking classes is never assumed to preserve fibres*: existence of lifts is proved for maps, and injectivity of q_* is proved from a mere pointed equality of projections, through the uniqueness of lifts up to mere equality (Lemma 5). Lifts exist merely, and any two lifts with equal pointed projections determine the same pointed homotopy class (Theorem 6); connectedness of the full lift spaces (pairs $(L, \alpha : L; q = F)$) is neither claimed nor needed.

1. Conventions

As in `doc/fibre-h3.pdf`: `isTrunc n X` is "X is $(n-2)$-truncated", `isConn n X` is "X is $(n-2)$-connected"; $K_m = K(\mathbb{Z}, m)$, $K_m^A = K(A, m)$ (`EMMaps.KK`); $F; G$ is `ptcomp F G`, with $(F; G)_1 = G_1 \circ F_1$ by definition; 0 is `ptconst`; $[F]$ is `hcl F`, $a; b$ on classes is `compC` with $[F]; [G] = [F; G]$ (`comp-beta`); equality of classes is mere equality of pointed maps (`effective`, `relates`). For a triple (A, B, k), $k : K_2^A \to_* W$ with $W = K_4^B$, $w_0 = *$; $X = \mathrm{Fib}k = (y : K_2^A) \times (k\, y = w_0)$ pointed by $(*, k_0)$, and $q = \mathrm{fpr} : X \to_* K_2^A$, $q(y, h) = y$, pointed by refl. $P = \mathrm{Fib}\kappa = (x : K_2) \times T_x$, $T_x = (\kappa x = *)$, $p_0 = (*, \kappa_0)$, $p = \mathrm{fpr} : P \to_* K_2$.

Levels used: K_2^A is isConn 3 and isTrunc 4; T_x is isConn 4 (`conn-id`, inlined in `cp`); W is isConn 5, so ΩW is isConn 4, $\Omega^2 W$ is isConn 3, $\Omega^3 W$ is isConn 2 (`conn-id`, `conn-down`); X is isConn 3 (`Triple.conn3`), so ΩX is isConn 2.

2. The lifting lemma

Fix a pointed type (Q, q_0) and a triple (A, B, k). Write $q_*(c) = c; [q]$ for $c \in [Q, X]_*$.

Date: 28 October 2026.

Lemma 1 (pointing irrelevance; `FibreH3.pi-mere`)**.** *If $\Omega(Z, z_0)$ is connected, pointed maps $(k, \zeta), (k', \zeta')$ with $k = k'$ are merely equal.*

Lemma 2 (lifts exist; `Fibres.Lift`)**.** *If $\theta : F; k = 0$ for $F : Q \to_* K_2^A$, there is $L : Q \to_* X$ with $L; q = F$ (`lift`, `lift-fst`).*

Lemma 3 (lifts are merely unique, unpointed)**.** *Assume (U_0): every $d : Q \to \Omega W$ is merely equal to $\lambda_.\mathsf{refl}$. Let $u : Q \to X$, $g' : Q \to K_2^A$, $h' : (x : Q) \to k(g'x) = w_0$ and $e : (\lambda x.\,(u\,x)_1) = g'$. Then merely $u = \lambda x.(g'x, h'x)$.*

Proof. J on e (g' and h' free). For $g' \equiv \lambda x.(u\,x)_1$, let $d\,x = (u\,x)_2^{-1} \cdot h'x : w_0 = w_0$. By ($U_0$), merely $\lambda_.\mathsf{refl} = d$. Then for each x

$$(u\,x)_2 \;\equiv\; (u\,x)_2 \cdot \mathsf{refl} \;=\; (u\,x)_2 \cdot d\,x \;=\; (u\,x)_2 \cdot ((u\,x)_2^{-1} \cdot h'x) \;=\; h'x$$

(`trans-cancel'`), so $u\,x = ((u\,x)_1, h'x)$, and $u = \lambda x.(g'x, h'x)$ by function extensionality. □

Lemma 4 ((U_0) from (U))**.** *Assume (U): every $d : Q \to_* \Omega W$ is merely equal to 0. Then (U_0) holds.*

Proof. Let $d : Q \to \Omega W$. Since ΩW is connected, merely $r : d\,q_0 = \mathsf{refl}$ (`conn-path`). By (U), merely $0 = (d, r)$; its first component gives $\lambda_.\mathsf{refl} = d$. □

Lemma 5 (lifts are merely unique)**.** *Assume (U). If $L, L' : Q \to_* X$ and $L; q = L'; q$, then merely $L = L'$.*

Proof. The first component of $L; q = L'; q$ is $e : (\lambda x.(L_1 x)_1) = (\lambda x.(L'_1 x)_1)$ (by definition of `ptcomp` and `fpr`). Lemma 3 with $u = L_1$, $g' = \lambda x.(L'_1 x)_1$, $h' = \lambda x.(L'_1 x)_2$ gives merely $L_1 = L'_1$ (Σ-η); Lemma 1 (ΩX connected) gives merely $L = L'$. □

Theorem 6 (the lifting lemma)**.** *Assume (E): for every $F : Q \to_* K_2^A$, merely $F; k = 0$; and (U). Then $q_* : [Q, X]_* \to [Q, K_2^A]_*$ is a bijection.*

Proof. `Bijection.Bij`: the target is a set; it suffices that q_* is injective and that each fibre is merely inhabited. *Surjective.* By induction on classes (into propositions), for $[F]$: by (E) merely $\theta : F; k = 0$; Lemma 2 gives L with $q_*[L] = [L; q] = [F]$. *Injective.* By induction on both classes (the statement is a proposition: classes form a set), for $[L], [L']$ with $[L; q] = [L'; q]$: merely $L; q = L'; q$ (`effective`), so merely $L = L'$ (Lemma 5) and $[L] = [L']$ (`relates`). □

So the lift space over each F (pairs $(L,\ L; q = F)$) is merely inhabited, and any two of its points have merely equal first components; nothing is claimed about its higher structure.

3. The source P

Lemma 7 (fibres of a first projection)**.** *For $B : A \to U$ and $a : A$, the fibre of $\mathrm{pr}_1 : \Sigma_A B \to A$ over a is equivalent to $B\,a$, by $((a', b), \pi) \mapsto \pi_* b$ with inverse $b \mapsto ((a, b), \mathsf{refl})$ (J on π). Hence the fibres of $p_1 : P \to K_2$ are $\mathsf{isConn}\,4$ (`equiv-conn`).*

Lemma 8 (p^* on classes into K_2^A)**.** *$p^* : [K_2, K_2^A]_* \to [P, K_2^A]_*$, $c \mapsto [p]; c$, is a bijection. More precisely: for $F : P \to_* K_2^A$ let $G_F = (\bar{F}, \zeta_F)$ with $\bar{F} = \mathrm{ext}\,F_1$ (`ConnectedMaps.Elim`, $m = 4$, along p_1, constant family K_2^A) and $\zeta_F = (\bar{F}(*) = F_1 p_0) \cdot F_0$ (`ext-beta` at p_0, since $p_1 p_0 \equiv *$). Then merely $p; G_F = F$.*

Proof. $(p; G_F)_1 = \lambda w.\bar{F}(w_1) = F_1$ by `ext-beta` and function extensionality; Lemma 1 (ΩK_2^A connected) gives merely $p; G_F = F$. This is surjectivity (by induction on classes). Injectivity: from $[p; G] = [p; G']$, merely $p; G = p; G'$; the first components give $G_1 \circ p_1 = G'_1 \circ p_1$, so $G_1 = G'_1$ (`Elim.unique`), and merely $G = G'$ by Lemma 1. `Bij` concludes. □

Lemma 9 (the fibre sequence)**.** *Merely* $p;\kappa = 0$.

Proof. $(p;\kappa)_1 = \lambda w.\kappa(w_1)$ and $\lambda w.\, w_2$ is a homotopy to $\lambda_.*$; Lemma 1 (ΩK_4 connected). □

Proposition 10 ((E) for P)**.** *For every triple* (A, B, k) *and* $F : P \to_* K_2^A$*, merely* $F; k = 0$.

Proof. Let $G = G_F$ (Lemma 8). By surjectivity of Φ_B (H4), for $g = \Phi_B^{-1}[G; k]$, $[\kappa; K(g, 4)] = [G; k]$, so merely $\kappa; K(g, 4) = G; k$. Then, merely,

$$F; k = (p; G); k = p; (G; k) = p; (\kappa; K(g, 4)) = (p; \kappa); K(g, 4) = 0; K(g, 4) = 0$$

(Lemma 8, `ptcomp-assoc`, Lemma 9, `comp-const-l`). □

Proposition 11 ((U) for P)**.** *Every* $d : P \to_* \Omega W$ *is merely* 0*:* ***`FibAt.merely-zero`*** *(**`doc/fibre-h3.pdf`**, Theorem 10, before the transfer along* e_3*).*

Corollary 12. *For every triple,* $q_* : [P, \mathrm{Fib}k]_* \to [P, K_2^A]_*$ *is a bijection.*

4. The source S^2

Let $(S^2, \mathsf{base}, \mathsf{surf})$ satisfy the sphere axiom and $\mathrm{ev} = \texttt{evPt} : \mathrm{Map}_*(S^2, Z) \simeq \Omega^2 Z$ (`pointed-up`), $\mathrm{ev}(F; G) = (\Omega^2 G)(\mathrm{ev}F)$ (`evPt-nat`).

Lemma 13. *If* $\Omega^2(Z, z_0)$ *is connected, any two* $F, F' : S^2 \to_* Z$ *are merely equal: merely* $\mathrm{ev}F = \mathrm{ev}F'$*, and* ev *is injective.*

Corollary 14. *(E) and (U) hold for* $Q = S^2$ *(*$Z = W$*:* $\Omega^2 W$ *is connected;* $Z = \Omega W$*:* $\Omega^3 W$ *is connected; for (E),* $F; k$ *and* 0 *are two maps* $S^2 \to_* W$*). So* $q_* : [S^2, \mathrm{Fib}k]_* \to [S^2, K_2^A]_*$ *is a bijection for every triple; in particular* $p_* : [S^2, P]_* \to [S^2, K_2]_*$ *is one (the triple* $(\mathbb{Z}, \mathbb{Z}, \kappa)$*).*

Lemma 15 (looping is an equivalence; Corollary 11)**.** *For* Y *a 2-type,* $\Omega : \mathrm{Map}_*(K_2, Y) \to \mathrm{Map}_*(\Omega K_2, \Omega Y)$ *is an equivalence: every* ***`Unloop`*** f *is contractible (**`Over.cor11`**,* $c = 1$*), which gives an inverse* $f \mapsto$ *(centre of Unloop*f*)*$_1$*, with* Ω *of it* $= f$ *and, at* $f = \Omega G$*,* (G, refl) *equal to the centre.*

Lemma 16 (precomposition by a pointed equivalence)**.** *For* $e : (X', x_0') \simeq_* (Z, z_0)$*,* $G \mapsto e_1; G$ *is an equivalence* $\mathrm{Map}_*(Z, Y) \to \mathrm{Map}_*(X', Y)$ *(pointed-equivalence induction,* ***`ptEqInd`****; at the identity it is homotopic to the identity,* ***`ptcomp-idl`****).*

Definition 17 (σ)**.** Let $g : \Omega K_1$ be the generator $\mathsf{unband}(1)$ (`CircleProperty.Circle.g`; it is locked, so the instances for different parameters Y are identified by a lemma, `g-def`: both have $\mathsf{band}(g) = \mathsf{band}(g \cdot \mathsf{refl}) = 1 + \mathsf{band}(\mathsf{refl})$ by `band-g`, then `band-inj`), $e_1 : \Omega K_2 \simeq_* K_1$ (`CupSquare.e1`), e_1^{-1} the map of its inverse `pt-inv e1` (`CupSquare.e1inv`, by definition), and

$$c = (\Omega e_1^{-1})(g) \;:\; \Omega^2 K_2, \qquad \sigma = \mathrm{ev}^{-1}(c) : S^2 \to_* K_2, \quad \mathrm{ev}\,\sigma = c.$$

Proposition 18 (K_2 is the 2-truncation of S^2, pointed)**.** *For every 2-type* (Y, y_0)*,* $\sigma^* : \mathrm{Map}_*(K_2, Y) \to \mathrm{Map}_*(S^2, Y)$*,* $G \mapsto \sigma; G$*, is an equivalence.*

Proof. Let $\Psi(G) = \mathrm{ev}^\circ(e_1^{-1}; \Omega G)$, where $\mathrm{ev}^\circ : \mathrm{Map}_*(K_1, \Omega Y) \simeq \Omega^2 Y$, $F \mapsto (\Omega F)(g)$, is the circle property (`Circle[ΩY].pointed-up`). Ψ is a composite of three equivalences (Lemmas 15, 16, the circle property). And

$$\mathrm{ev}(\sigma; G) = (\Omega^2 G)(\mathrm{ev}\,\sigma) = (\Omega^2 G)((\Omega e_1^{-1})(g)) = (\Omega e_1^{-1}; \Omega^2 G)(g) = (\Omega(e_1^{-1}; \Omega G))(g) = \Psi(G)$$

(`evPt-nat`, Definition 17, definition of `ptcomp`, `loop-comp`). So $\mathrm{ev} \circ \sigma^* = \Psi$ and σ^* is an equivalence (2-out-of-3, `cancel-equiv`, after `E.homotopy`: the square commutes pointwise). □

Hence $[\sigma]^* : [K_2, Y]_* \to [S^2, Y]_*$ is a bijection (`pre-equiv`).

Remark 19 ($p \circ s$ represents 1). Definition 17 makes σ the generator: under $\Omega e_1 : \Omega^2 K_2 \simeq \Omega K_1$ and the band $\Omega K_1 \simeq \mathbb{Z}$, c goes to g and then to 1. The proof uses only Proposition 18, which is the property that characterises a generator; the value 1 is not used.

5. The mapping theorem for targets Fib$k$

Definition 20 (s)**.** $s = p_*^{-1}[\sigma] \in [S^2, P]_*$ (Corollary 14), so $s;[p] = [\sigma]$. A class, obtained without choice.

Proposition 21. *For every abelian A, $s^* : [P, K_2^A]_* \to [S^2, K_2^A]_*$ is a bijection.*

Proof. $s^* \circ p^* = [\sigma]^*$: $s;([p];d) = (s;[p]);d = [\sigma];d$ (`comp-assoc`). p^* is a bijection (Lemma 8), and so is $[\sigma]^*$ (Proposition 18, K_2^A a 2-type); so $s^* = [\sigma]^* \circ (p^*)^{-1}$ is one (`cancel-r`: if $g \circ f$ and f are equivalences, so is g). □

Theorem 22 (the mapping theorem for Fib$k$)**.** *For every triple (A, B, k), $s^* : [P, \mathrm{Fib}k]_* \to [S^2, \mathrm{Fib}k]_*$ is a bijection.*

Proof. The square

$$\begin{array}{ccc} [P, \mathrm{Fib}k]_* & \xrightarrow{q_*} & [P, K_2^A]_* \\ {\scriptstyle s^*}\downarrow & & \downarrow{\scriptstyle s^*} \\ [S^2, \mathrm{Fib}k]_* & \xrightarrow{q_*} & [S^2, K_2^A]_* \end{array}$$

commutes by `comp-assoc`: $(s;c);[q] = s;(c;[q])$. The top (Corollary 12), the right (Proposition 21) and the bottom (Corollary 14) are bijections; so the left is one (2-out-of-3, `cancel-equiv`). □

Corollary 23 (transfer)**.** *If* Fib$k \simeq_* (X, x_0)$ *for some triple, $s^* : [P, X]_* \to [S^2, X]_*$ is a bijection (`ptEqInd`). Since the statement is a proposition, a* mere *pointed equivalence suffices.*

Remark 24 (what remains for `MapThm`). `Pi3Sphere.Conditional.MapThm` asks for s^* bijective at every simply connected 3-type. `SphereUniversal.Assemble` uses it at $X = P$ (Theorem 22 with the triple $(\mathbb{Z}, \mathbb{Z}, \kappa)$) and at $X = T = \|S^2\|3$; by Corollary 23, the latter follows from a mere Postnikov presentation $T \simeq_*$ Fib$k$ (§3d, not done here).

6. Correspondence with the formal proof

Status: checked (`./segc sag/pi3-s2-postnikov all`: 70 modules, 0 failed; a fresh process: 3003 declarations, 0 failed, 1.2 s; every `MapFib` declaration under 0.03 s; `mapfib.opens`, 110 entries, every line "nothing else opened"). Module `MapFib`: `fib-fst-ret`, `fib-fst` (Lemma 7); `cancel-snd` (the step of Lemma 3); `cancel-r`; `LevW[B]`: `cOW`, `cOOW`, `cOOOW` (levels); `Lifting[Q,q0,A,B,k]`: `Ex`, `Un`, `U0` ((E), (U), (U_0)), `qs` (q_*), `cOF`, `u0` (Lemma 4), `uniq0` (Lemma 3), `uniq` (Lemma 5), `Main[ex,un]`: `surj`, `inj-at`, `inj`, `equiv` (Theorem 6); `SourceP[`κ`,h4k]`: `cp` (Lemma 7), `PStar[A]`: `EA`, `cOA`, `ext-pt` (G_F), `ext-eq`, `pstar`, `surj`, `inj-at`, `inj`, `equiv` (Lemma 8); `cOK4`, `fseq` (Lemma 9); `ExP[A,B,k,F]`: `g`, `g-eq`, `chain`, `null`; `ex-P` (Proposition 10); `un-P` (Proposition 11); `qP` (Corollary 12); `pre-pteq` (Lemma 16); `SourceS[S2,base,surf,ump]`: `s2-paths` (Lemma 13), `ex-S`, `un-S`, `qS` (Corollary 14), `LoopEq[Y,y0,tY]`: `cell`, `equiv` (Lemma 15), `gen-c`, `sigma`, `sigma-ev`, `g-def` (Definition 17), `Sig[Y,y0,tY]`: `Psi`, `psi-equiv`, `sigma-nat`, `sigma-star`, `sigma-cls` (Proposition 18); `Final[S2,base,surf,ump,`κ`,h4k]`: `s`, `s-eq` (Definition 20), `right` (Proposition 21), `rep-fib` (Theorem 22), `rep-pteq` (Corollary 23 for a pointed equivalence; the mere version, `TC.rec` into the proposition `isEquiv`, is not written); `rep-cup`: Theorem 22 for `CupSquare.kappa`, `H4Proof.h4`.